

⇒ Ex 2.5

Reduce the following matrices into echelon form using elementary row operations.

→ echelon form:

$$\begin{bmatrix} 1 & A_{12} & A_{13} \\ 0 & 1 & A_{23} \\ 0 & 0 & 1 \end{bmatrix}$$

$$\textcircled{1} \begin{bmatrix} 5 & 9 & 3 \\ -3 & 5 & 6 \\ -1 & -5 & -3 \end{bmatrix} \begin{matrix} R_1 \\ R_2 \\ R_3 \end{matrix}$$

1. Now we can interchange any Row.

2. Note we can multiply any Number with any R.
ex → NR_n

$$\sim 3. R_i + 2R_j$$

(Initial Row add or subtract with final Row.

→ Mean jis Row ko liye hum ya jis row par hum operation perform kary gai us ko pehly likhy gai.

sol

By or interchange R_3 to R_1

$$\begin{bmatrix} 5 & 9 & 3 \\ -3 & 5 & 6 \\ -1 & -5 & -3 \end{bmatrix} \quad R_3 \leftrightarrow R_1$$

$$5 \begin{bmatrix} -1 & -5 & -3 \\ -3 & 5 & 6 \\ 5 & 9 & 3 \end{bmatrix} \rightarrow (-1) R_1$$

$$\begin{bmatrix} 1 & 5 & 3 \\ -3 & 5 & 6 \\ 5 & 9 & 3 \end{bmatrix}$$

Now

$$R_2 + 3R_1 \rightarrow R_2$$

$$\begin{bmatrix} 1 & 5 & 3 \\ 0 & 20 & 15 \\ 5 & 9 & 3 \end{bmatrix} \quad \begin{array}{l} -3 + 3(1) = 0 \rightarrow R_2 \\ 5 + 3(5) = 20 \rightarrow R_2 \\ 6 + 3(3) = 9 \rightarrow R_2 \end{array}$$

Now for $R_3 - 5R_1 \rightarrow R_3$.

$$\begin{bmatrix} 1 & 5 & 3 \\ 0 & 20 & 15 \\ 0 & 16 & -12 \end{bmatrix}$$

Now

$$\frac{1}{20} R_2$$

$$\begin{bmatrix} 1 & 5 & 3 \\ 0 & 1 & 3/4 \\ 0 & -16 & -12 \end{bmatrix} \quad \text{Now, } R_3 + 16R_2$$

$$\begin{bmatrix} 1 & 5 & 3 \\ 0 & 1 & 3/4 \\ 0 & 0 & 0 \end{bmatrix}$$

come with zero, then it is also
as echlon form. if last row consider